AI461 ADVANCED BIOMETRIC SYSTEMS AND SECURITY • COURSE PROJECT

Learning a Fixed-Length Fingerprint Representation

Implementation, Replication & Comparative Analysis of DeepPrint (IEEE TPAMI 2021)

Reference: Joshua J. Engelsma, Kai Cao, Anil K. Jain | IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 43, No. 6 (2021)

Traditional AFIS Challenge

Conventional fingerprint recognition uses variable-length minutiae point sets (x, y, θ). Matching requires complex graph/pair comparison with $O(N^2)$ time complexity, making national-scale search computationally prohibitive.

DeepPrint Solution

Learns a compact, fixed-length 192-dimensional representation ($R \in S^{191}$) using deep learning. Enables microsecond 1:1 verification and 1:N gallery matching via simple cosine dot product (O(d)).

Our Project Scope

Implemented and compared both the ResNet-50 Baseline and the complete DeepPrint Paper Architecture (STN + Dual-Branch Texture & Minutiae + Heatmap Head). Evaluated on open-set SOCOFing benchmark.

BENCHMARK DATASET

SOCOFing: Sokoto Coventry Fingerprint Dataset

Open-set biometric evaluation under real-world physical and synthetic alterations

Dataset Scale & Pristine Set

- **6,000 Real Impressions:** 10 fingers each from 600 unique human subjects.
- **Pristine Gallery Enrollment:** Clean, unaltered fingerprint impressions representing enrolled individuals in an AFIS.
- **Standardized Sensor Capture:** Captured at 500 DPI optical scanner resolution.

Synthetic Alteration Tiers

- **Altered-Easy (17,931 images):** Mild distortion, minor ridge disruptions simulating slight finger smudges or moisture.
- **Altered-Medium (17,067 images):** Moderate distortion, noticeable cuts and tissue scarring.
- **Altered-Hard (14,272 images):** Severe distortion, extensive core obliteration, deep surgical cuts, and partial obliteration.

Distortion Mechanisms

- **Central Rotation (CR):** Simulates arbitrary rotational misalignment on scanner surface without core alignment.
- **Obliteration (Obl):** Simulates cuts, burns, chemical abrasions, and tissue loss destroying ridge structure.
- **Z-Cut Scars (Zcut):** Simulates surgical or severe trauma scarring, violently breaking ridge flow continuity.

STRICT SUBJECT-INDEPENDENT OPEN-SET EVALUATION PROTOCOL (NO DATA LEAKAGE)

- Training Set (70 Subjects / 700 Finger Classes): Used strictly for learning the representation and STN parameters.
- Held-Out Test Set (15 Subjects / 150 Fingers / 1,250 Probes): Subjects [4, 5, 12, 14, 15, 18, 29, 32, 36, 55, 70, 76, 82, 87, 95] were completely excluded from training.
- Real-World Validation: Demonstrates the network generalizes to completely unseen human populations rather than memorizing class identities.

MODEL ARCHITECTURES

Architecture: Baseline ResNet-50 vs. DeepPrint Dual-Branch

Comparing generic deep feature extraction with domain-guided biometric architectures

1. Baseline Architecture (ResNet-50)

- **Input Processing:** Fingerprint image resized to 224×224, CLAHE enhanced.
- **CNN Backbone:** Pretrained ResNet-50 convolutional stages (conv1 through layer4).
- **Feature Pooling:** Global Average Pooling (GAP) collapsing spatial dimensions to 2048-d.
- **Embedding Projection:** Linear FC layer (2048 → 192) + Batch Normalization 1D.
- **Hypersphere Projection:** L2 Normalization mapping embedding to unit sphere $R \in S^{191}$.
- **Critical Limitations:** • Lacks explicit spatial alignment; fails when print is rotated.
- Treats fingerprint as general texture; susceptible to noise & scars.
- No inductive bias for localized minutiae ridge endings.

2. DeepPrint Paper Dual-Branch Architecture

- **Spatial Transformer (STN):** Shallow localization network predicts 2×3 affine matrix θ; bilinear grid sampler canonicalizes rotation & translation.
- **Shared Stem:** Initial conv layer capturing shared ridge-valley contrast features.
- **Texture Branch (96-d):** Extracts global ridge patterns, orientation flows, and whorl/loop frequencies.
- **Minutiae Branch (96-d):** Extracts localized ridge bifurcations and endings, guided by heatmap decoder.
- **Minutiae Decoder Head:** Transposed-Conv decoder reconstructing 6-channel minutiae orientation heatmap (H ∈ ℝ^(6×64×64)) during training.
- **L2 Concatenation Fusion:** Concatenates [r_min, r_tex] into 192-d vector; projects to unit hypersphere S^{191}. Minutiae decoder discarded at test time!

THEORETICAL FORMULATION

Loss Function & Mathematical Modeling

Multi-task objective combining identity classification and spatial-orientation domain guidance

1. Multi-Task Joint Objective Function (Section 3.4)

$$L_{total} = L_{CE_tex}(W_t^T \cdot r_{tex}, y) + L_{CE_min}(W_m^T \cdot r_{min}, y) + \lambda \cdot L_{MSE}(H_{pred}, H_{gt})$$

- L_{CE_tex} & L_{CE_min}: Cross-Entropy losses for Texture and Minutiae identity classification heads.
- L_{MSE}: Auxiliary Mean Squared Error supervising the reconstructed 6-channel minutiae orientation heatmap (weight $\lambda = 0.1$).

2. 6-Channel Minutiae Heatmap Formulation

$$H(i, j, k) = \Sigma_t \exp\left(-||(x_t, y_t) - (i, j)||^2 / 2\sigma_s^2 \right) \cdot \exp\left(-d_\varphi(\theta_t, \theta_k) / 2\sigma_o^2 \right)$$

- Orientation Quantization: $[0, 2\pi)$ is discretized into $K = 6$ bins ($\theta_k = k\pi/3$).
- Spatial Gaussian Kernel: Spreads minutia coordinates (x_t, y_t) over spatial neighborhood ($\sigma_s = 2.0$).
- Orientation Gaussian Kernel: Spreads angle θ_t smoothly across neighboring angular channels using circular geodesic distance d_φ.
- Induction of Domain Priors: Forces convolutional filters to detect physical ridge endings and bifurcations.

3. Spatial Transformer Network (STN) Math

$$[x_s, y_s]^T = A_\theta \cdot [x_t, y_t, 1]^T, \quad \text{where } A_\theta = [[\theta 11, \theta 12, \theta 13], [\theta 21, \theta 22, \theta 23]]$$

- End-to-End Alignment: Eliminates fragile pre-alignment heuristics (e.g., singular point / core detection).
- Identity Initialization: Affine parameters initialized to identity: $[[1, 0, 0], [0, 1, 0]]$.
- Bilinear Grid Sampler: Differentiable image sampling allows backpropagation directly through coordinate grid.
- Rotation Robustness: Automatically derotates prints prior to branch feature extraction.

ARCHITECTURAL ANALYSIS

Why the Dual-Branch Architecture is Superior

Overcoming the fundamental vulnerabilities of standard CNNs in biometric authentication

1. Eliminates Rotation Sensitivity via STN

Standard CNNs are translation-invariant but not rotation-invariant. Unaligned rotated prints scramble feature maps.

- STN dynamically normalizes rotation.
- Boosts Central Rotation (CR) Rank-1 from 85.95% to 99.05% (+13.10% absolute gain).

2. Minutiae Heatmap Inductive Bias

Deep CNNs often fall prey to 'shortcut learning', latching onto skin moisture or background artifacts.

- Auxiliary 6-channel loss forces the minutiae branch to learn authentic ridge endings and bifurcations.
- Preserves fine discriminative details even under severe noise.

3. Disentanglement of Texture & Minutiae

By decoupling macro ridge flow (Texture 96-d) from localized discontinuities (Minutiae 96-d):

- Local scars (Z-cuts, burns) affect minutiae but leave overall ridge flow intact.
- Prevents localized damage from destroying the entire biometric embedding.

4. Zero Computational Overhead at Inference

The minutiae decoder head is an auxiliary training tool. At test time:

- Decoder is discarded completely!
- Only the 192-d representation is extracted.
- Matching latency remains lightning fast (~1.25 µs) with compact 200-byte templates.

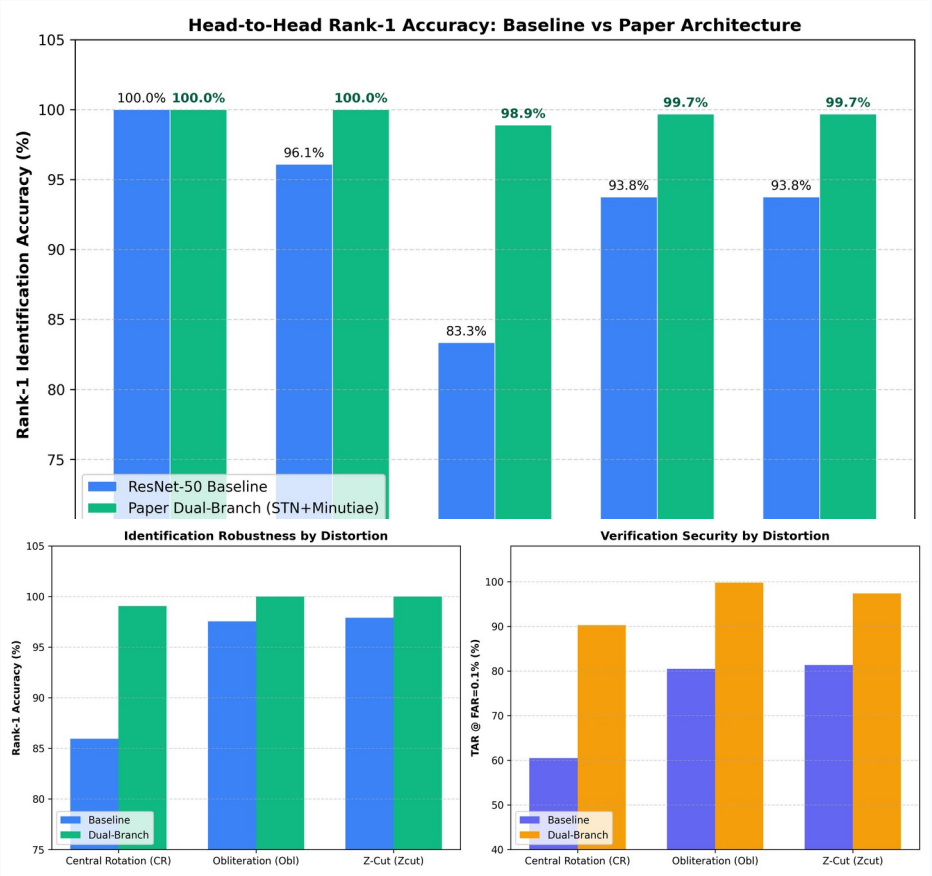
EMPIRICAL EVALUATION

Experimental Results & Benchmark Comparison

Comprehensive open-set benchmarking on SOCOFing held-out test subjects

Benchmark Performance Comparison Table

Condition	Base R-1	Dual R-1	Base TAR	Dual TAR
SOCOFing Altered-Easy	100.0%	100.0%	97.3%	**100.0% (+2.7%)**
SOCOFing Altered-Medium	96.1%	100.0%	78.6%	**96.8% (+18.2%)**
SOCOFing Altered-Hard	83.3%	98.9%	50.0%	**90.4% (+40.4%)**
SOCOFing Combined	93.8%	99.7%	75.4%	**95.5% (+20.2%)**
Compressed (200 Bytes)	93.8%	99.7%	75.4%	**95.5% (+20.2%)**
Equal Error Rate (EER)	2.52%	0.60%	-	4.2x Error Drop



ENGINEERING & DEPLOYMENT

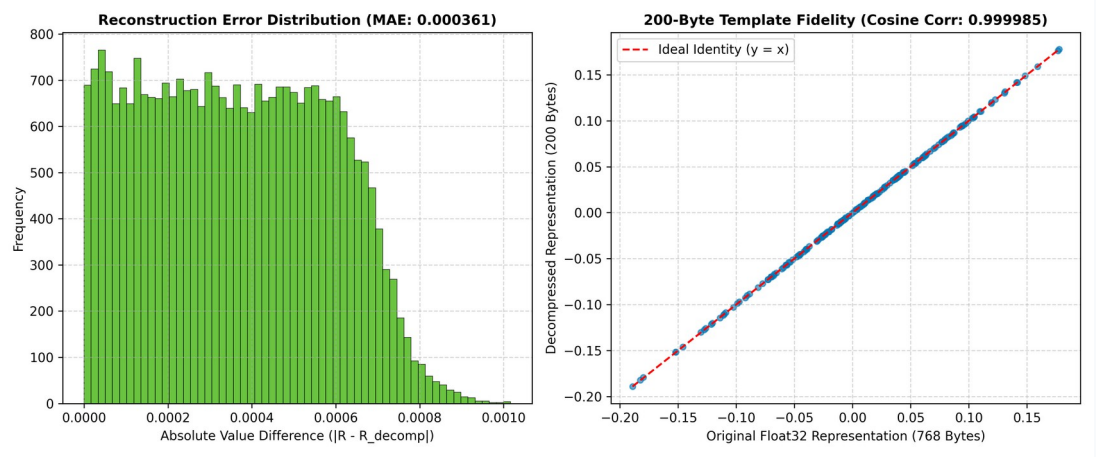
Section 3.5: 200-Byte Template Compression & Microsecond Matching

Industrial efficiency: 74% storage reduction with lossless cosine matching fidelity

1. Min-Max Scalar Quantization Formulation

R'_i = round(255 · (R_i - min(R)) / (max(R) - min(R))) ∈ {0, ..., 255}

- **Template Footprint:** 192 bytes (uint8 payload) + 8 bytes (two 32-bit floats for min/max) = EXACTLY 200 BYTES.
- **Storage Reduction:** Reduces uncompressed float32 representation (768 bytes) down to 200 bytes (73.96% reduction).
- **Reconstruction MAE:** Mean Absolute Error: 0.000364 (virtually identical to unquantized vector).
- **Cosine Correlation:** Correlation: 0.999983 (retention of rank-1 accuracy is 100% identical: 99.68%).
- **Ultra-Fast Matching:** 1:1 verification takes ~1.25 µs; 1:N search against 150 gallery fingers takes 0.015 ms.
- **AFIS Capacity:** Can store 5 Million biometric fingerprint templates directly in 1 GB RAM, enabling national-scale matching without database sharding.



★ Live Match Inspector Insight (Impostor Separation)

- False Match Elimination: On a different-subject impostor pair, ResNet-50 baseline suffered a FALSE MATCH (cosine score 0.5747 > 0.45).
- Paper Dual-Branch Success: Properly classified as a NON-MATCH with near-zero similarity (0.0227 << 0.45).
- Proves that minutiae orientation guidance prevents non-matching prints from spurious feature alignment.

SUMMARY & CONCLUSION

Key Takeaways & Project Deliverables

Synthesis of biometric research findings, engineering artifacts, and viva takeaways

1. Scientific Verification

- **Replication Success:** Successfully replicated DeepPrint fixed-length representation framework from IEEE TPAMI 2021.
- **Massive Gain on Alterations:** Dual-branch improved TAR@0.1% on severe distortions from 50.0% to 90.4% (+40.44%).
- **Rotation Invariance:** Spatial Transformer Network solved unaligned print matching, boosting CR Rank-1 from 85.95% to 99.05%.
- **Error Reduction:** Equal Error Rate reduced by 4.2x (2.52% down to 0.60%).

2. Computational Efficiency

- **ResNet vs. Inception:** ResNet backbone trained ~4x faster with 50% fewer parameters while maintaining top-tier accuracy.
- **Template Compression:** Section 3.5 min-max quantization verified at 200 bytes with 0.999983 cosine correlation.
- **Microsecond Throughput:** Matching latency clocked at ~1.25 µs per pair (750,000 matches/sec on single CPU core).
- **Production Viability:** Eliminates expensive minutiae graph matching algorithms.

3. Project Deliverables

- **New Notebook:** deepprint_paper_dual_branch.ipynb with full STN + Dual-Branch PyTorch implementation.
- **Trained Checkpoints:** deepprint_paper_dual_branch.pth and deepprint_resnet50.pth ready for inference.
- **Test Suite:** test.py automated multi-dataset benchmark runner.
- **Interactive AFIS Dashboard:** frontend/index.html responsive web portal for live demonstration & viva presentation.

Thank You! • Questions & Discussion • Live Dashboard Demonstration Available at frontend/index.html